

Preface

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Chapter 1

Homotopy Theory and the Fundamental Group

Definition 1.0.1. If f and g are continuous maps from topological spaces X and Y , we say that f is homotopic to g if there exists a continuous map $F : X \times [0, 1] \rightarrow Y$ such that:

$$F(x, 0) = f(x) \text{ and } F(x, 1) = g(x)$$

for all $x \in X$. We say F is a homotopy between f and g , and write $f \cong g$. If f is homotopic to a constant map, we say f is nullhomotopic.

Example 1.0.1. Let $f, g : X \rightarrow \mathbb{R}^n$ be continuous. Define:

$$F(x, t) := (1 - t)f(x) + tg(x).$$

It is easy to see that F is continuous, $F(x, 0) = f(x)$, and $F(x, 1) = g(x)$. The map F is called the *straight line homotopy*, because it moves the point $f(x)$ to the point $g(x)$ along a straight line.

Example 1.0.2. Define:

$$\begin{aligned} i : S^1 &\rightarrow \mathbb{R}^2 \setminus \{(0, 0)\}, \quad i(x) = x; \\ f : S^1 &\rightarrow \mathbb{R}^2 \setminus \{(0, 0)\}, \quad f(x) = (1, 0). \end{aligned}$$

Then i and f are *not* homotopic.

Definition 1.0.2. A path along a topological space X is a continuous function $f : [0, 1] \rightarrow X$. We call $f(0)$ the initial point and $f(1)$ the terminal point.

Definition 1.0.3. Two paths $f, g : [0, 1] \rightarrow X$ are path homotopic if their initial and terminal points are equal and there exists a continuous map $\bar{F} : [0, 1] \times [0, 1] \rightarrow X$ such that:

$$\begin{aligned} \bar{F}(s, 0) &= f(s) \quad \text{and} \quad \bar{F}(s, 1) = g(s) \\ \bar{F}(0, t) &= x_0 \quad \text{and} \quad \bar{F}(1, t) = x_1 \end{aligned}$$